

Tomas Ordonez Gonzalez

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EDUCATION

University of Alberta

Bachelor of Science with Specialization, Major in Computing Science

GPA: 3.9/4.0

Sep 2022 – Present

Edmonton, AB

EXPERIENCE

Research Assistant

Department of Computing Science, University of Alberta

May 2025 – Aug 2025

Edmonton, AB

- Awarded NSERC Undergraduate Student Research Award (USRA) to conduct research under Dr. Guohui Lin.
- Researched star packing approximation algorithms in directed graphs.
- Applied algorithm design, mathematical reasoning, and problem solving in collaboration with supervisor.
- Reviewed and presented relevant academic papers to the research group to assist with investigations.

Lead Lab Design Teaching Assistant

Department of Computing Science, University of Alberta

Jul 2024 – Dec 2024; Jul 2025 – Dec 2025

Edmonton, AB

- Worked in a team to design 20 large programming exercises in C for 1,300+ undergraduate students.
- Used Git and GitHub for version control and project management.
- Developed automated marking scripts in Bash and Python to evaluate student submissions efficiently.
- Assisted students with C programming, emphasizing memory management, data structures, and algorithms.

PROJECTS

ECS Physics Game - SKKS | C++, GLSL, ECS, SFML, Git, GitHub

- 2D physics based puzzle game built using a custom ECS game engine.
- Worked in a team using Git and GitHub for version control.
- Wrote various shader programs for interesting visual effects.
- Used batch sprite rendering for performance optimization.
- Implemented boid flocking algorithm to create realistic swarms of enemies.

Social Media Android Application - Y | Java, Firebase, Android Studio, Git, GitHub

- Collaborated in a team to develop a full-stack social media application for Android.
- Designed and implemented backend features to manage posts, user profiles, and follow relationships.
- Integrated Firebase Firestore for real-time data storage and synchronization across users.
- Contributed to frontend development, including navigation, image caching, and UI elements.
- Implemented offline functionality with local caching and sync mechanisms for seamless user experience.

Graph Methods Visualizer | TypeScript, JavaScript, HTML, CSS, Git

- Developed a web-based tool for creating and visualizing different types of graphs.
- Implemented interactive visualizations of graph methods such as BFS, DFS, Dijkstra, etc. using TypeScript.
- Added import/export functionality and built-in graph presets for intuitive user experience.

Ray Tracing Quaternion Julia Sets | C++, OpenGL, GLSL, Dear ImGui, Git

- Developed a graphics application used to explore the Quaternion Julia fractals in real time.
- Implemented physically based rendering techniques using GLSL shaders for efficient GPU rendering.
- Added immediate mode GUI in C++, providing full control over rendering and fractal settings.

SKILLS

Languages: English (fluent), Spanish (fluent))

Programming Languages: C++, C, Java, Python, TypeScript, JavaScript, SQL, Bash, Lisp, MATLAB, RISC-V Assembly

Markup and Styling Languages: HTML, CSS, Sass, LaTeX

Frameworks: React, Express JS

Databases: Firebase Firestore, MySQL, SQLite, MongoDB

Developer Tools: Git, GitHub, Linux, Docker, VS Code, Android Studio, Firebase

Libraries: OpenGL, Dear ImGui, Pandas, NumPy, Matplotlib, Ollama

Coursework: Numerical Methods, Algorithmics, Supervised Learning, Reinforcement Learning, Graph Theory